
Static Noise Margin Analysis of a 6T SRAM Cell

Read and Write Stability Optimization Using Cell Ratio and Pull-up Ratio Sweeps

Project summary

This report evaluates the stability of a conventional 6T SRAM bit-cell by varying the cell ratio (CR) and pull-up ratio (PR). Read SNM is extracted from read butterfly curves, while write SNM is obtained from write butterfly curves.

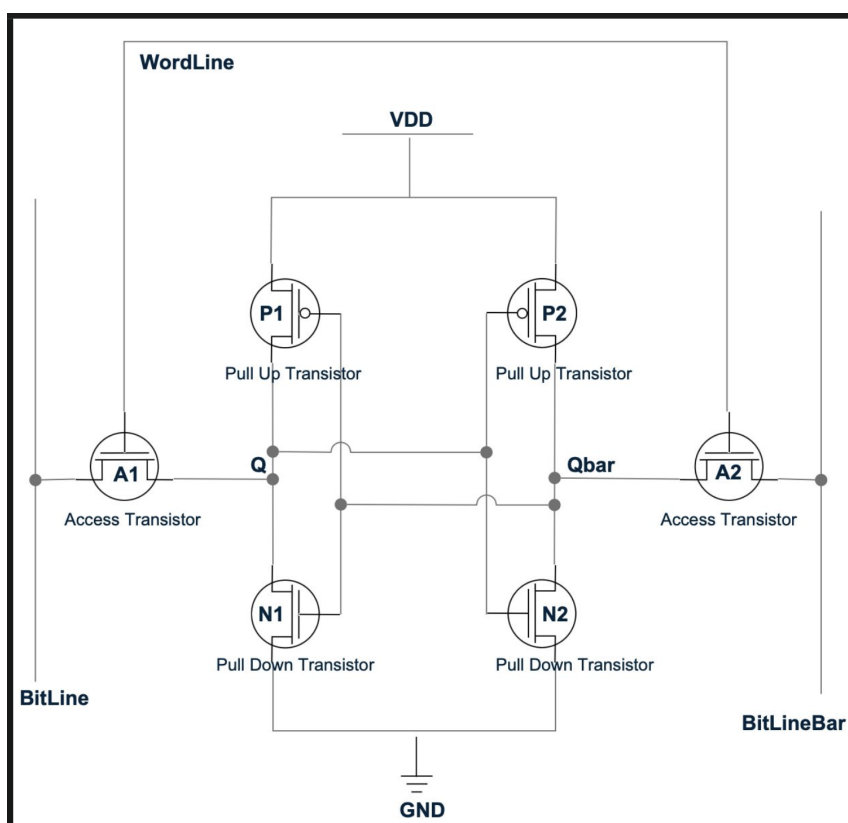


Figure 1: Transistor level 6T SRAM cell.

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Simulation platform	Cadence Virtuoso
Supply voltage	1.8 V
Technology	180 nm CMOS Technology

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Objective

The objective is to quantify how transistor sizing affects 6T SRAM static stability. The study examines:

- **Read stability:** effect of cell ratio (CR) on read static noise margin (RSNM).
- **Write ability:** effect of pull-up ratio (PR) on write static noise margin (WSNM).
- Selection of a sizing region that balances read robustness and write functionality.

SRAM Cell and Simulation Setup

6T SRAM Bit-cell

A conventional 6T SRAM cell consists of two cross-coupled CMOS inverters and two NMOS access transistors. The internal nodes are denoted by Q and $\bar{Q}$; the access transistors connect the cell to bit-lines BL and BLB under word-line (WL) control.

Table 1: Sizing definitions used in this work.

Parameter	Definition	Expression
Cell ratio (CR)	Strength of pull-down NMOS relative to access NMOS; it primarily controls read disturb tolerance.	$CR = W_{PD}/W_{AX}$
Pull-up ratio (PR)	Strength of pull-up PMOS relative to access NMOS; it strongly affects write ability.	$PR = W_{PU}/W_{AX}$

Table 2: Simulation conditions.

Item	Value
EDA tool	Cadence Virtuoso / Spectre
Supply voltage	1.8 V
Process technology	CMOS technology node used in simulation
Temperature	$27^\circ C$
Analysis type	DC sweep for butterfly-curve extraction
SNM extraction	Side length of the maximum embedded square

Methodology

Read SNM

For read analysis, the bit-lines are precharged and the word-line is asserted. The voltage transfer characteristics of the two inverters are overlaid to form the read butterfly curve. RSNM is reported as the side length of the largest square embedded between the two curves.

Write SNM

For write analysis, complementary data are forced on BL and BLB while WL is asserted. The write butterfly curve is generated using the corresponding DC sweep, and WSNM is extracted from the largest embedded square.

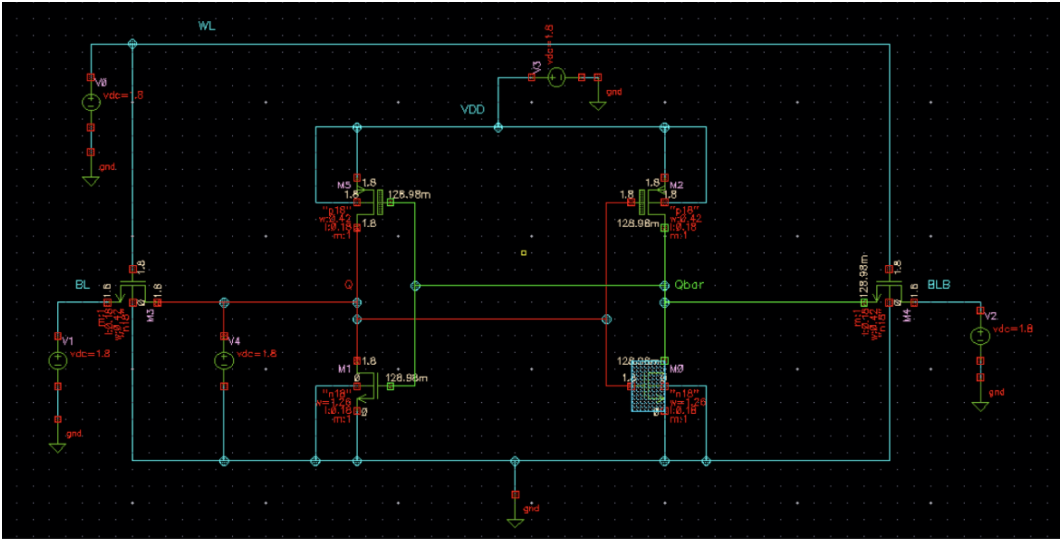


Figure 2: RSNM schematic.

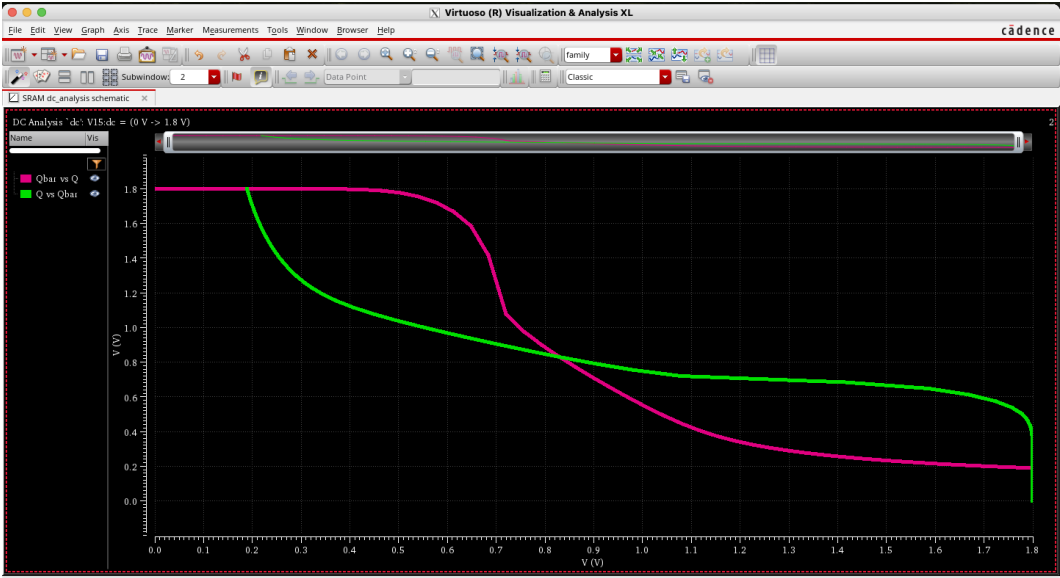


Figure 3: RSNM simulation in Cadence Virtuoso.

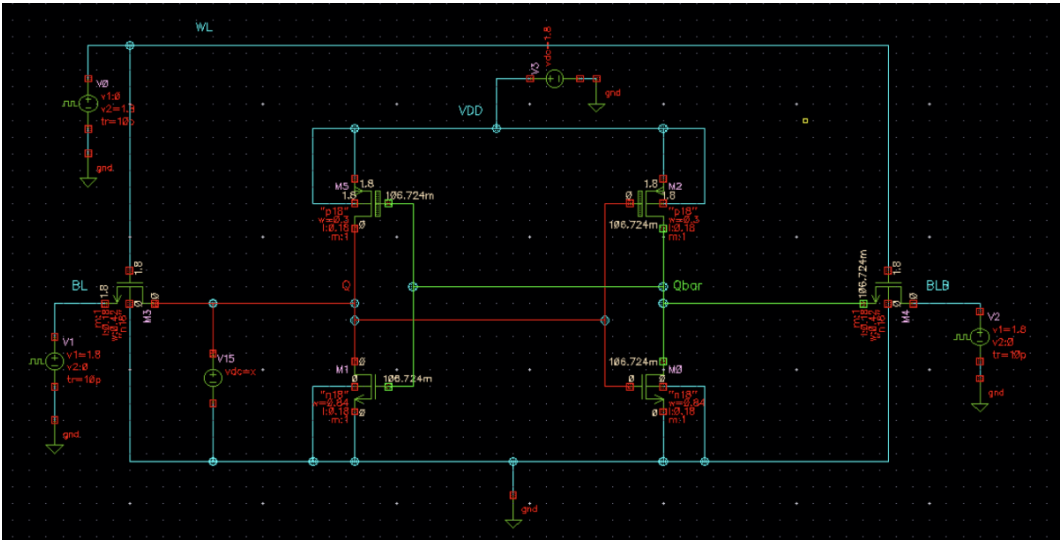


Figure 4: WSNM Schematic.

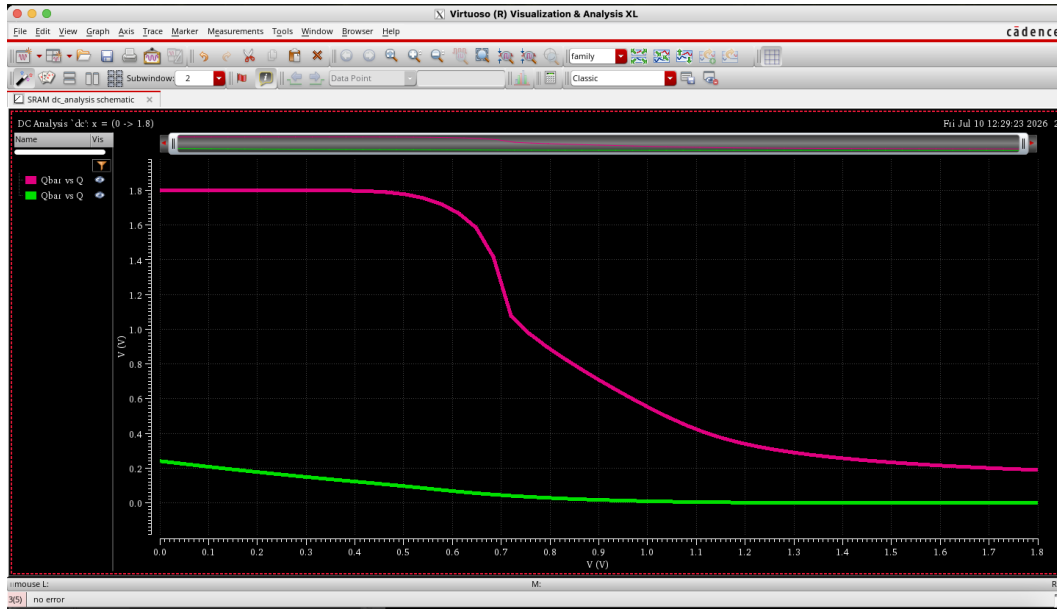


Figure 5: WSNM simulation in Cadence Virtuoso.

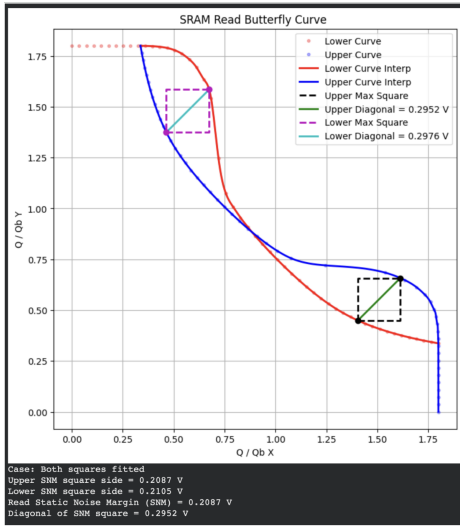
Results and Discussion

Read SNM versus Cell Ratio

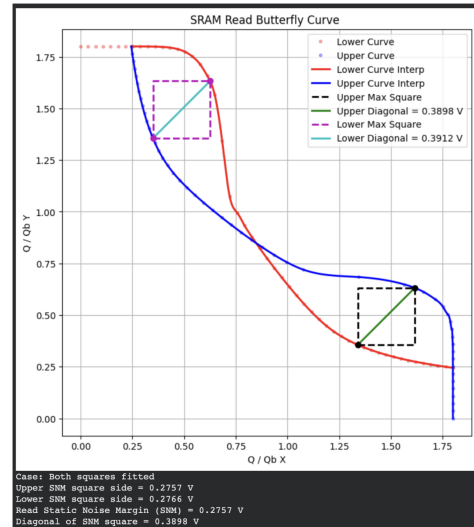
Table 3 summarizes the extracted RSNM values. The results show a monotonic increase in RSNM with CR, because a stronger pull-down device better preserves the stored logic ‘0’ during a read operation. We keep the access transistor’s width constant to standard 0.42 μm , and vary the pull down nmos width with respect to access transistor.

Table 3: Read stability results from the CR sweep.

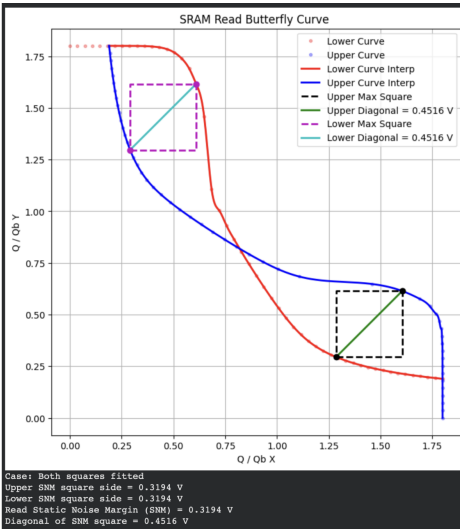
Cell Ratio (CR)	Pull-down NMOS width (μm)	RSNM (mV)
1.0	0.42	208.7
1.5	0.63	275.7
2.0	0.84	319.4
2.5	1.05	349.1
3.0	1.26	370.7



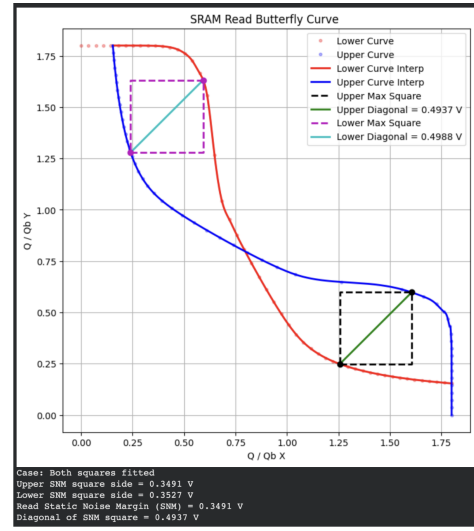
(a) Read butterfly curve at CR = 1.0.



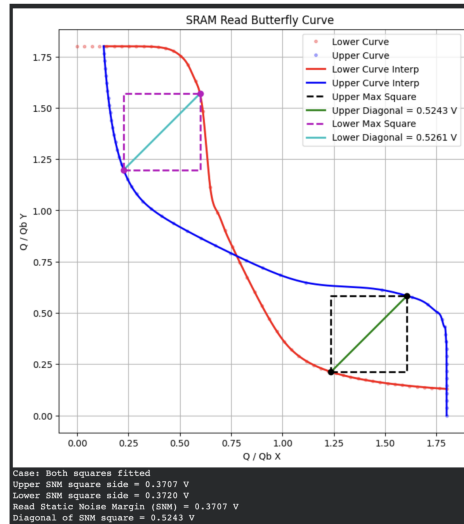
(b) Read butterfly curve at CR = 1.5.



(c) Read butterfly curve at CR = 2.0.



(d) Read butterfly curve at CR = 2.5.



(e) Read butterfly curve at CR = 3.

Figure 6: Read Static Noise Margins (RSNM) using butterfly curves for various values of CR.

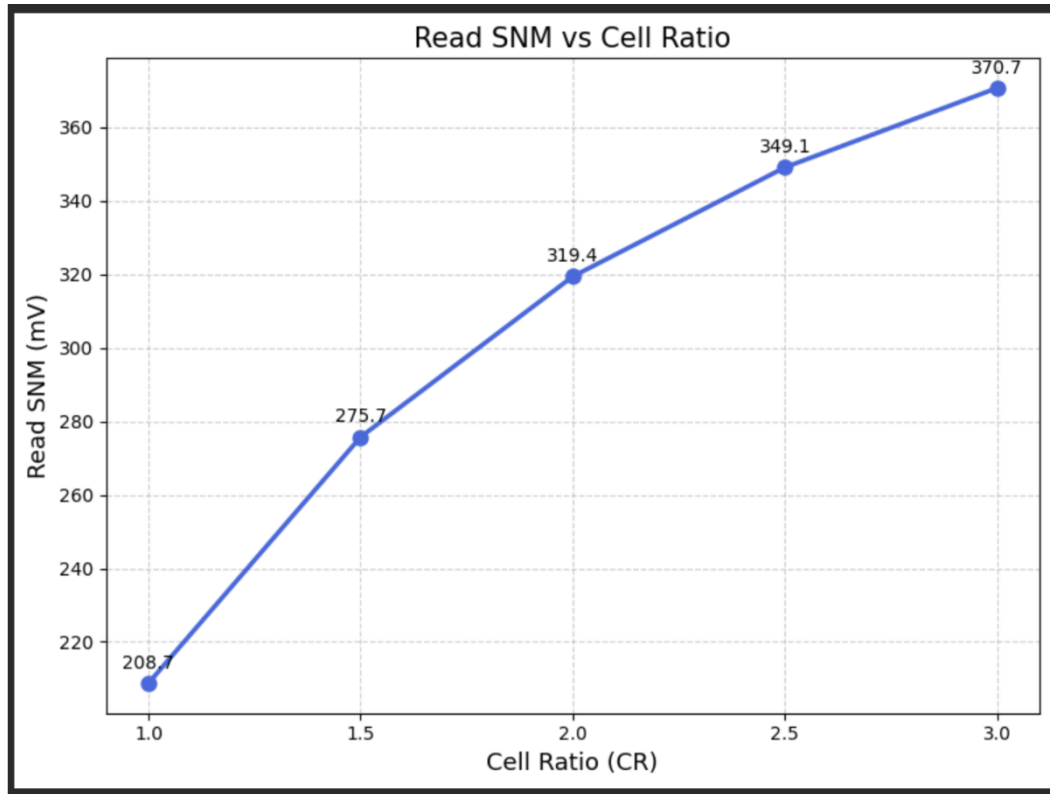


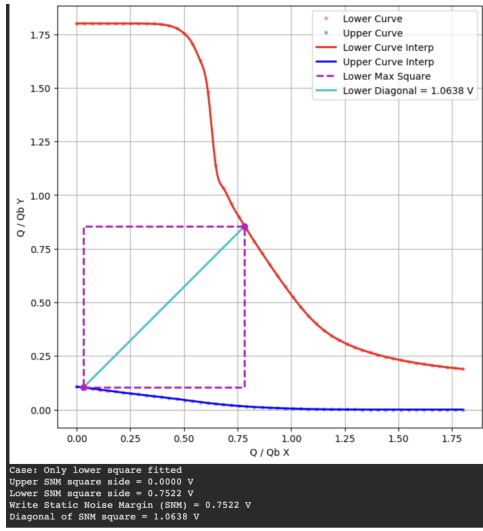
Figure 7: RSNM trend with CR.

Write SNM versus Pull-up Ratio

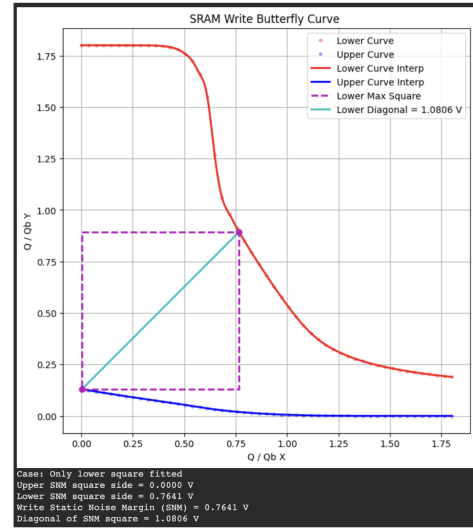
Table 4 summarizes the WSNM measurements. In this sweep, the highest measured WSNM is obtained at $PR = 0.71$. The variation is non-monotonic, indicating that write ability depends on the relative strengths of the pull-up PMOS and access NMOS rather than on PR alone. Also we have settled down the nmos widths for good read stability, hence we are getting non-monotonic behavior here. We keep the access transistor's width constant to standard $0.42\mu\text{m}$, and vary the pull up pmos width with respect to access transistor.

Table 4: Write stability results from the PR sweep.

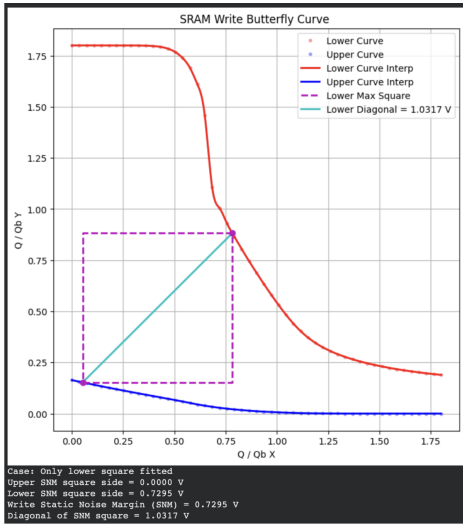
Pull-up Ratio (PR)	Pull-up PMOS width (μm)	WSNM (V)
0.52	0.22	0.7522
0.71	0.30	0.7641
1.00	0.42	0.7295
1.19	0.50	0.7511
1.43	0.60	0.7399



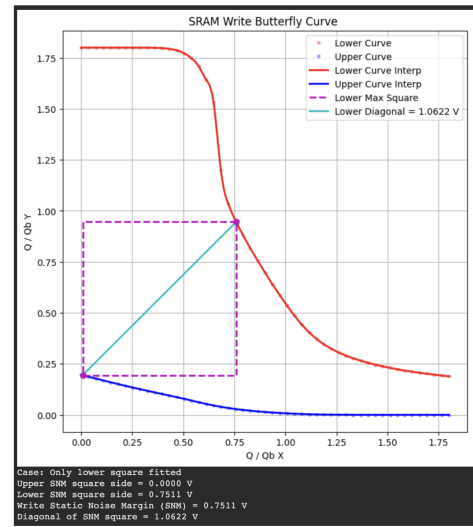
(a) Write butterfly curve at PR = 0.52.



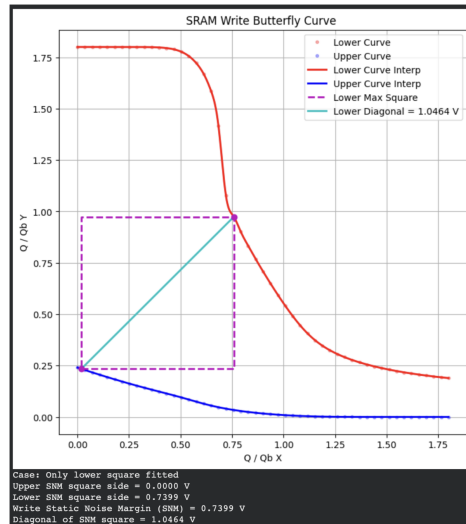
(b) Write butterfly curve at PR = 0.71.



(c) Write butterfly curve at PR = 1.00.



(d) Write butterfly curve at PR = 1.19.



(e) Write butterfly curve at PR = 1.43.

Figure 8: Write Static Noise Margins (WSNM) using butterfly curves for various values of PR.

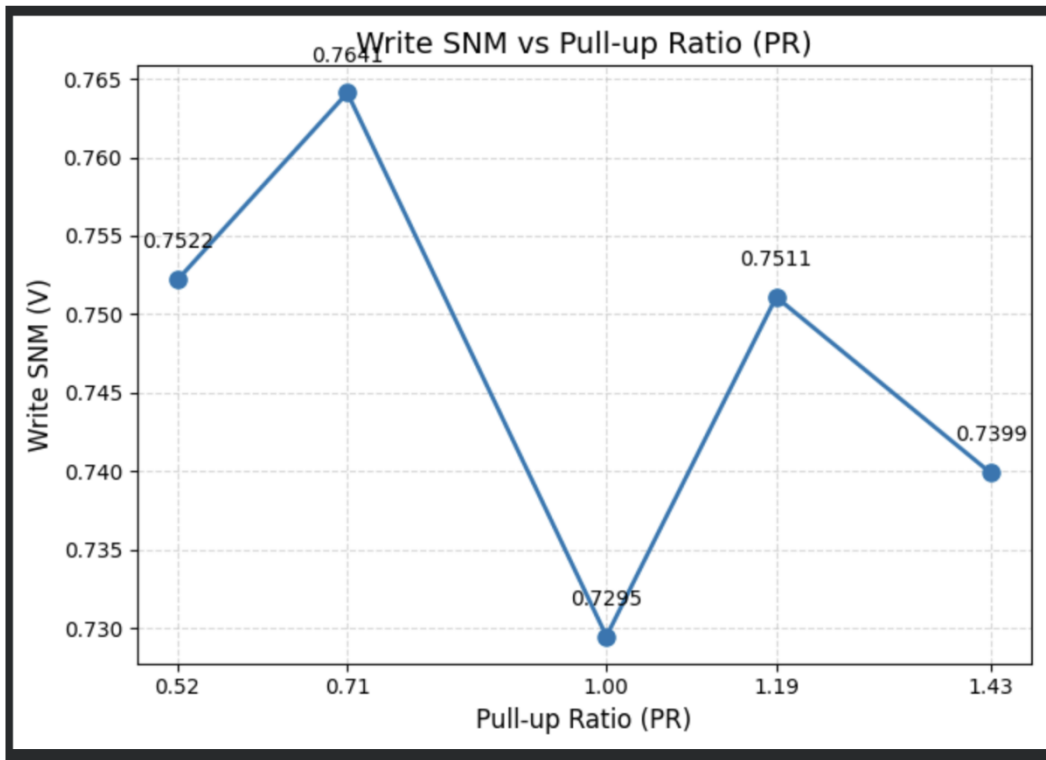


Figure 9: WSNM trend with PR.

Key Findings

- RSNM increased from **208.7 mV** at CR = 1.0 to **370.7 mV** at CR = 3.0, a gain of approximately **77.6%**.
- The maximum measured WSNM was **0.7641 V** at PR = 0.71.
- A larger CR improves read robustness; however, the final sizing choice should also account for area, leakage, delay, and write margin.
- The PR sweep demonstrates the need to optimize write ability jointly with read stability rather than maximizing a single transistor width.

Conclusion

A conventional 6T SRAM cell was successfully designed and characterized using Cadence Virtuoso in 180 nm CMOS technology. The read stability of the cell was evaluated by varying the Cell Ratio (CR), while the write stability was analyzed by varying the Pull-up Ratio (PR). The results show that the Read Static Noise Margin (RSNM) increases with increasing Cell Ratio, indicating improved read stability due to stronger pull-down NMOS transistors. The Write Static Noise Margin (WSNM) was found to be maximum at a Pull-up Ratio of approximately 0.71, demonstrating that an optimum PMOS sizing is required to achieve reliable write operation. Overall, the study highlights the importance of transistor sizing in balancing read stability and write ability, providing useful design guidelines for robust 6T SRAM cells.

Based on the obtained results, the final optimized transistor dimensions selected for the SRAM cell are an access NMOS width of **0.42 μm** , a pull-down NMOS width of **1.26 μm** (CR = 3), and a pull-up PMOS width of **0.30 μm** (PR = 0.71). These dimensions provide a balanced trade-off between read stability and write ability for the chosen technology.

The optimized SRAM cell obtained from this characterization will be used as the basic building block for the design and implementation of an SRAM array in future work.